

PHYSICS 7B – Spring 2025
Final Exam, C. Bordel
Tuesday, May 13th, 8-11 am

• Name:

• Student ID #:

• Section #:

- **Rules:** *You may work on this exam from 8:10-11:10 am PT. This midterm is closed book and closed notes and you are not allowed to use any electronic devices (computers, tablets, smartphones, calculators, ...). You are **not** allowed to communicate with any other students or discuss the content of the exam with anyone besides the Physics 7B teaching staff. Any violations of this policy will be considered a breach of academic integrity.*
- **Partial/Full Credit:** *Remember that you need to show your work and justify your answers in order to get full credit.*
- **Honors Statement:** *It is expected that during this examination, as with any examination that they undertake at the university, students adhere to the usual standards of academic integrity at the University of California at Berkeley as outlined on by the [Center for Teaching and Learning](#). Therefore, by submitting your exam, you are affirming the following statement:*
"I swear on my honor that I have neither given nor received aid on this exam. In addition, I abided by all of the examination policies as outlined above."

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Problem 1 - 25 points

n moles of a monoatomic ideal gas are taken from pressure P_A and volume V_A to pressure $P_B=4P_A$, following a reversible process described by $\frac{P}{T^2} = \text{constant}$.

- a-** Determine the final volume V_B . *Hint: it might be helpful to rewrite the equation describing the process in terms of P and V .*

- b-** Draw the thermodynamic process from state A to state B on a P - V diagram.

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- c- Determine the work done by the gas from A to B and comment on the sign.

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- d-** Determine the change in entropy of the gas from A to B . Does the result violate the second law of thermodynamics ? Explain. *Hint : the first law of thermodynamics might be useful.*

Problem 2 - 25 points

We consider a two-dimensional infinite sheet lying in the (xy) plane, with $z=0$ on the sheet. The sheet carries a uniform surface charge distribution such that $\sigma > 0$. We can show that the electric potential created at any point above or below the charged sheet satisfies:

$$V(x, y, z) = \frac{-\sigma}{2\epsilon_0} |z| + \text{constant}.$$

- a-** Using the $V(x, y, z)$ equation given above and assuming that we choose $V(z=0)=0$, determine the constant and plot V vs. z .
- b-** Based on the electric potential function, draw two equipotential surfaces on either side of the uniformly charged sheet, as well as some electric field lines. Explain.

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c- Using the electric potential function provided above, determine the electric field $\vec{E}(x, y, z)$ generated above and below the uniformly charged sheet.

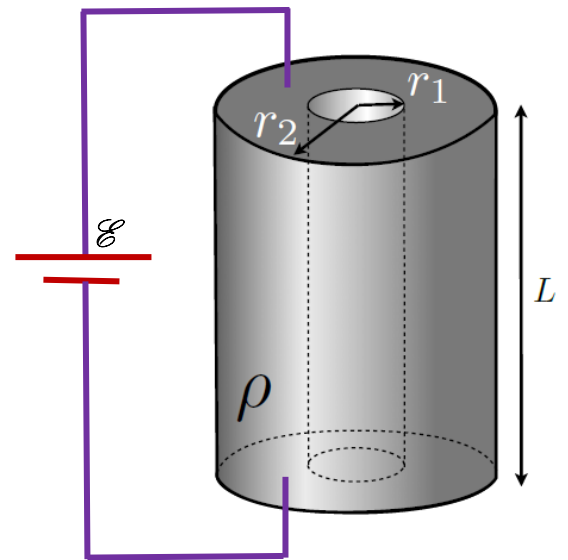
d- Using another method, determine the electric field generated by the uniformly charged sheet and compare with what you found in (c).

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Problem 3 - 25 points

Part 1

A cylindrical resistor made of a material of resistivity ρ , with inner radius r_1 , outer radius r_2 and length L , is connected to an ideal battery supplying an emf $\mathcal{E}$, as shown in the figure. The resistor is made of a conducting material of density d , atomic mass M_a , and has 2 free electrons per atom.



- a- Determine the resistance of the cylindrical resistor.

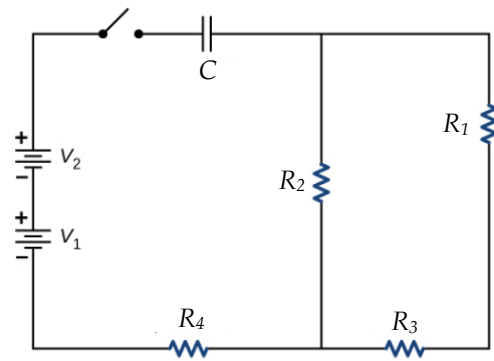
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- b-** Determine the drift velocity of the electrons in the resistor. *Hint: it might be helpful to determine the current density throughout the resistor.*

Part 2

Consider the following circuit, containing two batteries ($V_1=V_2=V_0$), four resistors ($R_1=R_2=R_3=R_4=R$) and one capacitor (C). The switch is initially open, and the capacitor uncharged. At time $t=0$, the switch is closed.

Express your answers in terms of V_0 , R and C .



- c- Determine the equivalent resistance and equivalent *emf* of the circuit.

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d- Determine at what time the current reaches 50% of its maximum value.

Problem 4 - 25 points

A very long and straight cylindrical wire of radius a carries some current with a non-uniform current density of magnitude $j(r) = j_0 \frac{a}{r}$, where j_0 is a positive constant and r the radial distance measured from the center of the wire.

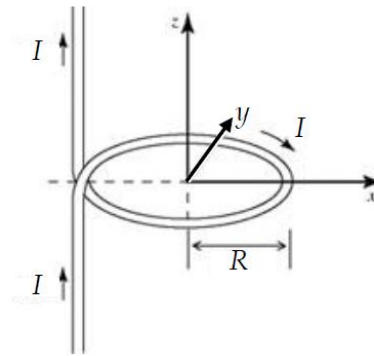
- a-** Draw a diagram showing the direction of the magnetic field created by the current-carrying wire, and explain the best choice of method to determine the magnitude of the magnetic field.

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- b-** Determine the magnitude of the magnetic field created by the current-carrying wire at any distance r from the center of the wire. Consider $r < a$ and $r \geq a$.

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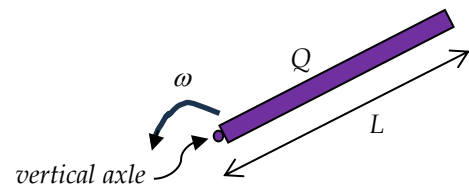
Now a section of the wire is bent into a circular loop of radius R (with $R \gg a$) lying in the horizontal xy -plane, while the long straight sections are parallel to the z -axis. The origin of the coordinate system is at the center of the loop. The $(\hat{x}, \hat{y}, \hat{z})$ basis is such that $\hat{x}$ points to the right, $\hat{y}$ points into the page, and $\hat{z}$ points upward.



- c- Using part (b), express the magnetic field $\vec{B}_1$ generated by the straight wire at the center of the loop, in vector form. You may use I in your answer.
- d- Determine the magnetic field $\vec{B}_2$ created by the loop at the origin and express the net magnetic field generated at the center of the loop. You may use I in your answer.

Problem 5 - 25 points

A uniformly charged rod of length L and charge Q ($Q>0$) is spun horizontally about a vertical axle mounted at one end, as shown in the top-view figure. The rod is mounted on frictionless bearings so it keeps spinning at constant angular speed ω in the counter-clockwise direction viewed from above after being pushed. You may consider a $\hat{z}$ unit vector pointing out of the page.



- a-** Determine, in terms of Q , ω , L and dr , the infinitesimal current dI generated by the motion of an infinitesimal charge dQ carried by an infinitesimal section located between r and $r+dr$ from the pivot point on the rod.
Hint: each section of the rod performs a full circle during one time period.

- b-** Determine the magnitude and direction of the magnetic dipole moment $\vec{\mu}$ of the spinning rod.

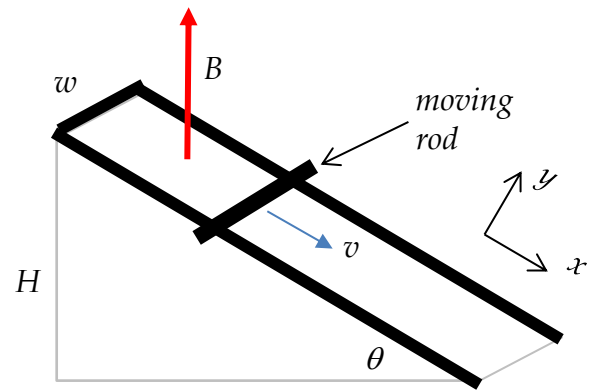
Name:

- c- Determine the direction and magnitude of the infinitesimal magnetic field created by the infinitesimal current considered in part (a), at a distance $z \gg L$ along the rotation axis of the rod.

- d- Determine the direction and magnitude of the net magnetic field created at distance $z \gg L$ as a function of $\vec{\mu}$ and z .

Problem 6 - 25 points

Three fixed conducting rails of negligible resistance are placed on the sides and at the top of a ramp. The ramp has a total height H , width w , and forms an angle θ with the horizontal direction. A conducting rod of mass m and resistance R is held by an experimentalist and slides down the ramp with constant speed v , as shown in the figure. A uniform magnetic field of magnitude B is applied vertically upward over the entire ramp. The moving rod is initially at the top of the ramp and starts sliding down at time $t=0$. You may neglect the effect of the induced magnetic field on the moving rod.



a- Determine the flux of magnetic field through the conducting loop as a function of time.

b- Determine the absolute value of the induced current that is generated in the conducting loop, and indicate its orientation on a diagram.

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- c- Determine the magnitude of the magnetic force acting on the moving rod and indicate its direction on a diagram.

Now instead of being maintained at constant speed, the rod is released from the top and slides down without friction. After speeding up, the rod reaches a constant speed called terminal speed.

- d- Determine the terminal speed of the rod.

Name:

Thermal expansion

$$\Delta l = \alpha l_0 \Delta T$$

$$\Delta A = 2\alpha A_0 \Delta T$$

$$\Delta V = \beta V_0 \Delta T = 3\alpha V_0 \Delta T$$

Calorimetry

$$Q = mc\Delta T = nC\Delta T \text{ (no phase transition)}$$

$$Q = mL \text{ (for a phase transition)}$$

$$\sum_{i=1}^n Q_i = 0 \text{ (for a closed system)}$$

Heat conduction

$$H = \frac{dQ}{dt} = -kA \frac{dT}{dx}$$

Ideal gases

$$PV = Nk_B T = nRT$$

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T$$

$$f_{\text{Maxwell}}(v) = 4\pi N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}}$$

$$\Delta E_{\text{int}} = Q - W$$

$$dE_{\text{int}} = dQ - PdV$$

$$W = \int PdV$$

$$E_{\text{int}} = \frac{d}{2} Nk_B T = nC_V T$$

$$C_V = \frac{d}{2} R$$

$$C_P - C_V = R = N_A k_B$$

$$\gamma = \frac{C_P}{C_V} = \frac{d+2}{d}$$

$$e = \frac{W_{\text{net}}}{Q_{\text{in}}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}}$$

$$e_{\text{ideal}} = 1 - \frac{T_L}{T_H}$$

$$S = \int \frac{dQ}{T} \text{ (For reversible processes)}$$

$$dQ = TdS$$

$$\Delta S_{\text{syst}} + \Delta S_{\text{env}} > 0 \text{ (For irreversible processes)}$$

Electrostatics

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\vec{F} = Q\vec{E}$$

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\lambda = \frac{dQ}{dl} ; \quad \sigma = \frac{dQ}{dA} ; \quad \rho = \frac{dQ}{dV}$$

$$\vec{p} = Q\vec{d}$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$U = -\vec{p} \cdot \vec{E}$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\Delta U = q\Delta V$$

$$V(b) - V(a) = - \int_a^b \vec{E} \cdot d\vec{l}$$

$$dV = \frac{dQ}{4\pi\epsilon_0 r}$$

$$\vec{E} = -\vec{\nabla}V$$

$$\vec{F} = -\vec{\nabla}U$$

Capacitors

$$Q = CV$$

$$C_{\text{eq}} = C_1 + C_2 \text{ (In parallel)}$$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} \text{ (In series)}$$

$$\epsilon = K\epsilon_0$$

$$C_D = KC_0$$

$$U = \frac{Q^2}{2C}$$

$$u = \frac{\epsilon}{2} |\vec{E}|^2 \text{ (energy density)}$$

	Q	W
Isoobaric ($P=const$)	$nC_p\Delta T$	$P\Delta V$
Isochoric ($V=const$)	$nC_v\Delta T$	0
Isothermal ($PV=const$)	$nRT \ln\left(\frac{V_f}{V_0}\right)$	$nRT \ln\left(\frac{V_f}{V_0}\right)$
Adiabatic ($PV^\gamma = const$)	0	$-\frac{d}{2}(P_f V_f - P_0 V_0)$

C_v, C_p and $\gamma = C_p/C_v$ for an Ideal Gas

1. For a monoatomic gas, $d=3$

$$C_v = \frac{3R}{2}, C_p = \frac{5R}{2} \text{ and } \gamma = \frac{5}{3} = 1.67$$

2. For a diatomic gas, $d=5$

$$C_v = \frac{5R}{2}, C_p = \frac{7R}{2} \text{ and } \gamma = \frac{7}{5} = 1.4$$

3. For a triatomic or polyatomic gas, $d=6$

$$C_v = 3R, C_p = 4R \text{ and } \gamma = \frac{4}{3} = 1.33$$

Current

$$I = \frac{dQ}{dt}$$

$$I = \int \vec{j} \cdot d\vec{A}$$

$$\vec{j} = nQ\vec{v}_d = \frac{\vec{E}}{\rho}$$

Resistance

$$R = \int \rho \frac{dl}{A}$$

$$\rho(T) = \rho(T_0)(1 + \alpha(T - T_0))$$

$$R_{eq} = R_1 + R_2 \text{ (In series)}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \text{ (In parallel)}$$

DC circuits

$$P = IV$$

$$V = IR \text{ (Ohm's law)}$$

$$\sum_{junc.} I_{in} = \sum_{junc.} I_{out} \text{ (junction rule)}$$

$$\sum_{loop} V_i = 0 \text{ (loop rule)}$$

$$\text{Charging capacitor: } \frac{dQ}{dt} + \frac{Q}{RC} = \frac{\mathcal{E}}{R}$$

$$Q(t) = C\mathcal{E}(1 - e^{-t/(RC)}) \text{ and } I(t) = \frac{dQ}{dt}$$

with $C\mathcal{E} = Q_{max}$ and $\tau = RC$

$$\text{Discharging capacitor: } \frac{dQ}{dt} + \frac{Q}{RC} = 0$$

$$Q(t) = C\mathcal{E}e^{-t/(RC)} \text{ and } I(t) = -\frac{dQ}{dt}$$

Magnetism

$$d\vec{F}_m = Id\vec{l} \times \vec{B}$$

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$\vec{\mu} = NI\vec{A}$$

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{encl}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \hat{r}}{r^2}$$

Integrals

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^\infty x^{2n} e^{-ax^2} dx = \frac{(2n)!}{n! 2^{2n+1}} \sqrt{\frac{\pi}{a^{2n+1}}}$$

$$\int_0^\infty x^{2n+1} e^{-ax^2} dx = \frac{n!}{2a^{n+1}}$$

$$\int (1+x^2)^{-1} dx = \arctan(x)$$

$$\int (1+x^2)^{-\frac{1}{2}} dx = \ln(\sqrt{1+x^2} + x)$$

$$\int x(1+x^2)^{-\frac{1}{2}} dx = \sqrt{1+x^2}$$

$$\int (1+x^2)^{-\frac{3}{2}} dx = \frac{x}{\sqrt{1+x^2}}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2)$$

$$\int \frac{1}{x} dx = \ln(x)$$

$$\int \frac{1}{\cos(x)} dx = \ln\left(\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right|\right)$$

$$\int \frac{1}{\sin(x)} dx = \ln\left(\left|\tan\left(\frac{x}{2}\right)\right|\right)$$

Electromagnetic Induction

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$M = N_1 \frac{\Phi_1}{I_2} = N_2 \frac{\Phi_2}{I_1}$$

$$L = N \frac{\Phi_B}{I}$$

$$U = \frac{1}{2} LI^2$$

$$U = \int \frac{1}{2\mu_0} |\vec{B}|^2 dV$$

Gradient and displacement

- *Cylindrical coordinates :*

$$\vec{\nabla} f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{\partial f}{\partial z} \hat{z}$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + dz \hat{z}$$

- *Spherical coordinates :*

$$\vec{\nabla} f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

Lengths, Areas and Volumes

$$\text{Arc length: } dL = R d\theta$$

$$\text{Cylinder: } A = 2\pi R H ; V = \pi R^2 H$$

$$\text{Cylindrical shell: } dV = 2\pi R H dR$$

$$\text{Thin Circular Ribbon : } dA = 2\pi R dR$$

$$\text{Section of Thin Ribbon : } dA = R dR d\theta$$

$$\text{Sphere: } A = 4\pi R^2 ; V = \frac{4}{3}\pi R^3$$

$$\text{Spherical shell: } dV = 4\pi R^2 dR$$

Name:

Taylor Expansions near x=0

$$\sin(x) \approx x$$

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

$$e^x \approx 1 + x + \frac{x^2}{2}$$

$$(1+x)^\alpha \approx 1 + \alpha x + \frac{(\alpha-1)\alpha}{2} x^2$$

$$\ln(1+x) \approx x - \frac{x^2}{2}$$

Motion and Energy

$$\Delta K + \Delta U = W_{nc}$$

$$K = \frac{1}{2}mv^2$$

$$\vec{F}_{sp} = -k \vec{x}; U_{sp} = \frac{1}{2}kx^2$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \text{ (simple harmonic motion)}$$

Trigonometry

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = 2\cos^2(x) - 1$$

$$\sin(a+b) = \sin(a) \cos(b)$$

$$+ \cos(a) \sin(b)$$

$$\cos(a+b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$c^2 = a^2 + b^2 - 2ab \cos(\theta) \text{ (law of cosine)}$$